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SUPERSEDING
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FEDERAL SPECIFICATION

Drill, Masonry, Rotary Driven

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 This specification covers one type of masonry drill, carbide tipped, core type, for use in drilling holes in plain or reinforced concrete and other type of masonry used with portable power rotary drills.

1.1.1 Federal specification coverage. This specification does not include all types, classes and sizes of masonry drills covered by the title of this specification, but includes only those most commonly used by the Federal Government.

2. APPLICABLE DOCUMENTS.

2.1 Government Publications. The issues of the following documents, in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

2.1.1 Federal Standards:
Fed. Std. No 123 – Marking for Domestic shipment (Civilian Agencies).

2.1.2 Military Standards:
MIL-STD-129 - Military Marking for Shipment and Storage

2.2 Other publications: American Society for Testing Materials Standard:

2.2.2. ASTM Designation E18 – Standard Test Methods for Rockwell Hardness of Metallic Materials.

(Single copies of Government publications required by activities outside the Federal Government for bidding purposes are available without charge from the Defense Logistics Agency "ASSIST" public site, **https://quicksearch.dla.mil/qsSearch.aspx.**) **(**Single copies of ASTM publications may be purchased at their website, **https://www.astm.org/)**

3. REQUIREMENTS.

3.1 Material. – The body of the drills shall be steel bar stock, or steel tubing suitable for its intended use. The shank of the drill shall be made from steel bar stock; the body of the drill shall have carbide tipped inserts for cutting (drilling) operation.

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3.2 Identification markings. – Each drill shall be stamped or otherwise permanently marked with its nominal size and the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.3 Design. – The drills shall have a right hand lead with spiral grooves (worm type), cut on the periphery of the body or may be furnished without spiral grooves with openings on the body, designed for the removal of material during drilling operation. The body shall be bored approximately 80 percent of its length and shall be provided with a slot or other openings for removal of core. The body shall be so designed with slots to receive the carbide-tip inserts which shall be properly brazed to permit efficient drilling of the material. The shank shall be concentric with the body within a maximum of 0.025 inch run out (total indicator reading). The outer edge of each carbide tip shall be in a circle, concentric with the body diameter at the drilling end. The design of the shank shall be straight round, hexagonal, Spline or SDS type shanks and of sufficient length to function efficiently in performing satisfactory work within its size range. An optional configuration of a removable shank may be allowed. The removable shank may be screwed directly into the drill body using a right hand rotation. Drills shall be furnished with a removable single carbide tipped pilot drill when specified. (See 6.1). The drills shall be similar to figure 1 and shall conform to table I for the nominal size specified (see 6.1).

3.3.1 Illustration. – The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of drills which are otherwise in accordance with the requirements of this specification.

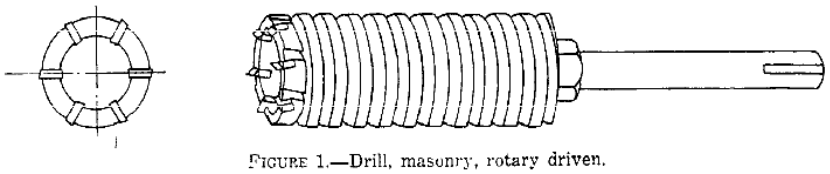


FIGURE 1.—Drill, masonry, rotary driven.

3.4 Hardness. – The carbide insert tips shall have a minimum hardness of 90.5 on the Rockwell "A" scale. The body of the drill shall be properly tempered and hardened to not less than 10 or more than 50 Rockwell "C" scale.

3.5 Finish. – Surfaces, other than carbide cutting inserts, shall be machined finish in accordance with good commercial shop practice.

3.6 Workmanship. – Workmanship shall be first class in every respect. Masonry drills and shanks shall be free from rust, laps, cracks, and other imperfections which may impair their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Place of inspection. – Inspection shall be performed at the place of manufacture.

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Table I. – Drills, masonry, carbide tipped, core type, rotary

Size	Body diameter		Carbide Tip diameter (*1)	Shank diameter or distance across flats +/- 0.010	Overall length min.
Nominal	Minimum	Maximum			
inches	inches	inches	inches	inches	inches
1/2	0.47	0.5	0.525	0.375	6
9/16	0.532	0.562	0.587	0.375	6
5/8	0.595	0.625	0.65	.375 or .5	8
5/8	0.595	0.625	0.65	0.5	6
11/16	0.657	0.687	0.712	.375 or .5	8
11/16	0.657	0.687	0.712	0.5	6
3/4	0.72	0.75	0.775	0.5	8
13/16	0.72	0.812	0.837	0.5	8
7/8	0.802	0.875	0.9	0.5	10
1	0.928	1	1.025	0.5	10
1-1/8	1.055	1.125	1.155	0.5	10
1-1/4	1.176	1.25	1.28	0.5	10
1-3/8	1.301	1.375	1.405	0.5	10
1-1/2	1.425	1.5	1.525	0.5	11
1-5/8	1.55	1.625	1.655	0.5	11
1-3/4	1.72	1.75	1.78	0.5	11
1-7/8	1.72	1.875	1.905	0.5	11
2	1.97	2	2.03	.5 or .75	11
2-1/8	2.095	2.125	2.157	0.75	12
2-1/4	2.22	2.25	2.281	0.75	15
2-3/8	2.345	2.375	2.406	0.75	15
2-1/2	2.47	2.5	2.531	0.75	15
2-5/8	2.595	2.625	2.656	0.75	15
2-3/4	2.72	2.75	2.781	0.75	15
2-7/8	2.845	2.875	2.906	0.75	15
3	2.97	3	3.312	.75 or 1	15
3-1/4	3.22	3.25	3.281	.75 or 1	15
3-1/2	3.47	3.5	3.531	.75 or 1	15
3-3/4	3.72	3.75	3.781	.75 or 1	15
4	3.97	4	4.031	.75 or 1	15

(*1)
Drills with a carbide tip diameter of up to 3/4-inch drill diameter shall have a tolerance of +0.01/ -0.00
Drills with a carbide tip diameter from 13/16 inch to 1-1/2 inches shall have a tolerance of +0.015/ -0
Drills with a carbide tip diameter 1-5/8 inches and above drill diameter shall have a tolerance of +0.031/ -0

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4.3 Test procedures.

4.3.1 Drilling test. – Masonry drills shall be tested on an upright drilling machine with positive feed, and operated at the speeds and feeds specified in table II. The minimum depth that drill is to be tested shall be 4 inches for a minimum of three holes for each size drill as specified in table II. Drilling tests shall be performed dry and without use of an air jet. When tested as specified in table II, there shall be no indication of excessive wear on body and on cutting edges of carbide inserts, no indication of bending, deforming or fractures in the body of the drill, no signs of spalling, chipping, or cracking of cutting edges of the inserts.

Table II. – Drilling Test

Size of drill (inches)	Revolutions per minute (rpm)	Inch feed per revolution
½ to ¾	300 to 400	0.008
13/16 to 1 7/8	200 to 300	0.008
2 to 4	100 to 200	0.008

4.3.1.1 Test block. – The test block shall be of molded masonry having a minimum thickness of four inches (the depth of one test hold). The block shall be thoroughly cured a minimum of five days in a steam box under controlled temperature and pressure conditions or shall be adequately moist cured for a minimum period of 28 days after stripping the block from the molds. Keeping the block buried in moist sand at room temperature (70 to 75 degrees Fahrenheit) or keeping in a fog room at 95 percent humidity and at room temperature shall be considered adequate moist curing. The following material shall be required for the test block:

(a) Two parts sharp washed sand (medium coarse).
(b) One part cement
(c) Sufficient water to make mixture of still consistency (approximately 5 ½ gallons to one 94-pound sack of cement).
(d) Two parts coarse, hard aggregate, 3/8 to ½ inch size. The aggregate shall be hard flint bluestone, “agate” pebbles, or other equivalent stone or rock which will lose not more than 50 percent weight if tested in accordance with test C-131, test ample grading “A”, of the “Standard Method of Test for Abrasion of Coarse Aggregate by use of the Los Angeles Machine” as specified in the current issue of the Book of American Society for Testing Materials (ASTM) Standards. (Crushed find grained igneous rock or stone, either basalt or diabase, shall be considered as equivalent aggregate provided it is of a quality which will meet the abrasion test requirements.)

The drill supplier shall furnish the inspector with written certification that the coarse aggregate used in the test block is of a quality which will meet the abrasion test requirements.

4.3.2 Concentricity test. The shank and body shall be tested for concentricity, using a V-block and dial indicator gage, or any other equally accurate method to determine compliance with 3.3. Drills having a run out greater than 0.025 inch (total indicator reading) shall be considered defective.

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5. MARKING.

5.1 Civil agencies. – marking for shipment shall be in accordance with the standard marking of Federal Standard No. 123 – Marking for Domestic shipment (Civilian Agencies).

5.2 Military. – marking for shipment shall be in accordance with MIL-STD-129 - Military Marking for Shipment and Storage.

6. NOTES.

6.1 Ordering data. – Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (nominal) and shank diameter (see table I).
(c) Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).
(d) Marking, if different.

Notice. – When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission or manufacture, use or sell any patented invention that may in any way be related thereto.

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Preparing Activity
GSA-FAS

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